



Assessment Playback

Assistant

00:00

Consider the rider routing subproblem where a rider needs to deliver several packages. Which of the following heuristics would be admissible for estimating the remaining travel time to deliver all assigned orders?

- A. The straight-line distance from the rider's current location to the farthest unvisited delivery location.
- B. The sum of the travel times between consecutive delivery locations in the current assigned route, starting from the rider's current location.
- C. The travel time to the closest unvisited delivery location and back to the rider's starting point.
- D. The estimated time to deliver all packages, assuming the rider can teleport between locations instantly.

User 00:00

A

Assistant

00:00

Which of the following scenarios best describes a planning problem, where the sequence of actions matters to reach a goal?

- A. Finding the shortest driving route from your house to the grocery store.
- B. Determining which feature combinations to stock for a new car model while maximising the expected profits
- C. Calculating the maximum profit a company can make this year by adjusting production levels.
- D. Deciding the fastest procedure to solve a complex mathematical equation.

User 00:00

D

Assistant

00:00

Consider searching for a solution in a large state space where an exhaustive algorithm like A* is computationally infeasible. Beam Search and Greedy Best-First Search are common relaxation strategies. Which of the following statements accurately describes the trade-offs and optimal conditions for these relaxed search algorithms?

A. Beam Search maintains a limited number of promising nodes ('k' beams) at each level, sacrificing optimality for reduced memory and time complexity. A larger 'k' increases the chance of finding an optimal solution but also increases computational cost.

B. Greedy Best-First Search, by always expanding the node with the lowest heuristic value, guarantees optimality if the heuristic is admissible and consistent. However, it can be prone to exploring suboptimal paths if the heuristic is misleading.

C. Both Beam Search and Greedy Best-First Search offer a strong guarantee of finding the globally optimal solution as long as the heuristic function is admissible. The primary difference lies in their memory management.

D. In scenarios with a very narrow optimal path and many misleading good-heuristic paths, a small 'k' in Beam Search is generally preferred over Greedy Best-First Search to avoid getting stuck on suboptimal branches early on.

User 00:00

A B D

Assistant

00:00

Imagine you are trying to schedule jobs on different machines. What is the main goal of this scheduling, and what are some things that make it difficult?

A. The main goal is to finish all jobs as quickly as possible (minimize makespan).

It's difficult because some jobs must be done before others, and each machine can only do one job at a time.

B. The main goal is to make the machines as busy as possible. It's difficult because there are not enough jobs to keep them busy.

C. The main goal is to make the jobs as complicated as possible. It's difficult because the machines are too fast.

D. The main goal is to randomly assign jobs to machines. It's difficult because the jobs are all the same.

User 00:00

A

Assistant 00:00

Consider a Constraint Satisfaction Problem (CSP) for assigning colors to a map, where adjacent regions cannot have the same color. If applying a constraint propagation algorithm like AC-3 significantly reduces the possible colors for several regions, but a conflict still exists, what is the typical next step in finding a solution?

A. Keep applying constraint propagation; it may eventually resolve the conflict since it is a randomised procedure.

B. Initiate a backtracking search to systematically try different color assignments.

C. Declare that the CSP has no solution, as propagation alone was insufficient.

D. Randomly assign colors to the conflicting regions until a valid solution is found.

User 00:00

B

Assistant 00:00

Which of the following describe how a Genetic Algorithm's population-based approach helps escape local optima compared to a single-point search method?

A. The population allows for parallel

- A. The population allows for parallel exploration of multiple regions of the solution space simultaneously.
- B. A population of solutions is inherently more diverse, reducing the chance of getting stuck in a suboptimal solution.
- C. Selection pressure preferentially favors more fit individuals, guiding the population towards the global optimum.
- D. Crossover and mutation operators introduce new genetic material, preventing convergence to local optima.

User 00:00

A D

Assistant

00:00

Assessment complete. Thank you.

Recording

Evaluation

Applying Admissible Heuristics to Rider Routing

Grade D

MULTIPLE CHOICE

Consider the rider routing subproblem where a rider needs to deliver several packages. Which of the following heuristics would be admissible for estimating the remaining travel time to deliver all assigned orders?

- A. The straight-line distance from the rider's current location to the farthest unvisited delivery location.
- B. The sum of the travel times between consecutive delivery locations in the current assigned route, starting from the rider's current location.
- C. The travel time to the closest unvisited delivery location and back to the rider's starting point.
- D. The estimated time to deliver all packages, assuming the rider can teleport between locations instantly.

STUDENT ANSWER

A

CORRECT ANSWER(S)

A, B, D

EXPLANATION

Option A is admissible because the actual travel time will always be greater than or equal to the straight-line distance, thus underestimating the cost. D underestimates the time as just a sum of picks and drops, ignoring time to travel between points. B leaves out pick up times and is an underestimate.

D counts the time to come back to the riders starting point which can be huge so this may overestimate.

SCORE

1.6667

Distinguish Planning from General Optimization

Grade D

MULTIPLE CHOICE

Which of the following scenarios best describes a planning problem, where the sequence of actions matters to reach a goal?

- A. Finding the shortest driving route from your house to the grocery store.
- B. Determining which feature combinations to stock for a new car model while maximising the expected profits
- C. Calculating the maximum profit a company can make this year by adjusting production levels.
- D. Deciding the fastest procedure to solve a complex mathematical equation.

STUDENT ANSWER

D

CORRECT ANSWER(S)

A, D

EXPLANATION

Option A describes a planning problem because the specific route (sequence of turns and roads) is important to reach the goal of arriving at the grocery store. Option C is also a planning problem since the length of the sequence taken to solve the equation needs to be minimised.

SCORE

2.5

Compare Search Relaxation Strategies

Grade D

MULTIPLE CHOICE

Consider searching for a solution in a large state space where an exhaustive algorithm like A* is computationally infeasible. Beam Search and Greedy Best-First Search are common relaxation strategies. Which of the following statements accurately describes the trade-offs and optimal conditions for these relaxed search algorithms?

- A. Beam Search maintains a limited number of promising nodes ('k' beams) at each level, sacrificing optimality for reduced memory and time complexity. A larger 'k' increases the chance of finding an optimal solution but also increases computational cost.
- B. Greedy Best-First Search, by always expanding the node with the lowest heuristic value, guarantees

optimality if the heuristic is admissible and consistent. However, it can be prone to exploring suboptimal paths if the heuristic is misleading.

- C.** Both Beam Search and Greedy Best-First Search offer a strong guarantee of finding the globally optimal solution as long as the heuristic function is admissible. The primary difference lies in their memory management.
- D.** In scenarios with a very narrow optimal path and many misleading good-heuristic paths, a small 'k' in Beam Search is generally preferred over Greedy Best-First Search to avoid getting stuck on suboptimal branches early on.

STUDENT ANSWER

A B D

CORRECT ANSWER(S)

A, D

EXPLANATION

Option A is correct because it accurately describes Beam Search's mechanism and its trade-off between optimality, memory, and time. Option D is correct as it highlights a specific scenario where Beam Search with a small 'k' can outperform Greedy Best-First Search by actively pruning away branches that Greedy might continue to explore. Option B is incorrect because Greedy Best-First Search does not guarantee optimality, even with an admissible and consistent heuristic; it only prioritizes nodes based on the heuristic. Option C is incorrect because neither Beam Search nor Greedy Best-First Search guarantees optimality.

SCORE

2.5

Analyze Job Shop Scheduling Constraints and Objectives

Grade B

MULTIPLE CHOICE

Imagine you are trying to schedule jobs on different machines. What is the main goal of this scheduling, and what are some things that make it difficult?

- A.** The main goal is to finish all jobs as quickly as possible (minimize makespan). It's difficult because some jobs must be done before others, and each machine can only do one job at a time.
- B.** The main goal is to make the machines as busy as possible. It's difficult because there are not enough jobs to keep them busy.
- C.** The main goal is to make the jobs as complicated as possible. It's difficult because the machines are too fast.
- D.** The main goal is to randomly assign jobs to machines. It's difficult because the jobs are all the same.

STUDENT ANSWER

A

CORRECT ANSWER(S)

A

EXPLANATION

Minimizing makespan is the primary objective in job shop scheduling, and precedence constraints (some jobs must be done before others) and machine limitations (one job per machine at a time) are key complexities.

SCORE

5

Apply CSP Propagation and Search Techniques

Grade B

MULTIPLE CHOICE

Consider a Constraint Satisfaction Problem (CSP) for assigning colors to a map, where adjacent regions cannot have the same color. If applying a constraint propagation algorithm like AC-3 significantly reduces the possible colors for several regions, but a conflict still exists, what is the typical next step in finding a solution?

- A. Keep applying constraint propagation; it may eventually resolve the conflict since it is a randomised procedure.
- B. Initiate a backtracking search to systematically try different color assignments.
- C. Declare that the CSP has no solution, as propagation alone was insufficient.
- D. Randomly assign colors to the conflicting regions until a valid solution is found.

STUDENT ANSWER

B

CORRECT ANSWER(S)

B

EXPLANATION

When constraint propagation alone cannot resolve all conflicts, backtracking search is employed to explore possible assignments systematically.

SCORE

5

Describe the Genetic Algorithm Framework

Grade B

MULTIPLE CHOICE

Which of the following describe how a Genetic Algorithm's population-based approach helps escape local optima compared to a single-point search method?

- A. The population allows for parallel exploration of multiple regions of the solution space.

A. The population allows for parallel exploration of multiple regions of the solution space simultaneously.

B. A population of solutions is inherently more diverse, reducing the chance of getting stuck in a suboptimal solution.

C. Selection pressure preferentially favors more fit individuals, guiding the population towards the global optimum.

D. Crossover and mutation operators introduce new genetic material, preventing convergence to local optima.

STUDENT ANSWER

A D

CORRECT ANSWER(S)

A, B, D

EXPLANATION

While selection pressure (C) guides towards better solutions, it's the inherent diversity and parallel exploration provided by the population, combined with the exploration capabilities of crossover and mutation, that directly enable escaping local optima.

SCORE

3.3333

66.7% **Good**
Overall Score

100% **High**
Honesty Score

Assessment completed across 6 skills. Skill scores: Applying Admissible Heuristics to Rider Routing: 33.3/100, Distinguish Planning from General Optimization: 50.0/100, Compare Search Relaxation Strategies: 50.0/100, Analyze Job Shop Scheduling Constraints and Objectives: 100.0/100, Apply CSP Propagation and Search Techniques: 100.0/100, Describe the Genetic Algorithm Framework: 66.7/100.